Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- jqassistant
- Neo4j Python Driver

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	58384	52.135088
1	[Git, Change, Create]	17242	15.396567
2	[Git, Commit]	11255	10.050363
3	[Git, Change, Delete]	8758	7.820620
4	[File, Git]	5891	5.260479
5	[Git, Change, Rename]	3354	2.995017
6	[Git, Tag]	1732	1.546622
7	[Author, Git, Person]	1265	1.129605
8	[Json, Key]	668	0.596503
9	[Json, Value, Scalar]	603	0.538460
10	[Committer, Git, Person]	370	0.330398
11	[NPM, Dependency]	338	0.301823
12	[Type, TS, Primitive]	291	0.259854
13	[Type, TS, Declared]	277	0.247352
14	[TS, ExternalDeclaration]	215	0.191988
15	[Git, Change, Copy]	138	0.123230
16	[Type, TS, Literal]	136	0.121444
17	[Json, Value, Object]	133	0.118765
18	[Type, TS, Union]	119	0.106263
19	[Type, TS, ObjectMember]	102	0.091083
20	[NPM, Script]	91	0.081260
21	[TS, Property]	65	0.058043
22	[TS, Function]	47	0.041970
23	[Git, Branch]	43	0.038398
24	[Type, TS, FunctionParameter]	40	0.035719
25	[Type, Object, TS]	39	0.034826
26	[File, Directory]	34	0.030361
27	[Type, TS, Function]	34	0.030361
28	[TS, Parameter]	33	0.029468
29	[Package, File, Json, NPM]	29	0.025896

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

```
<Figure size 640x480 with 0 Axes>
```

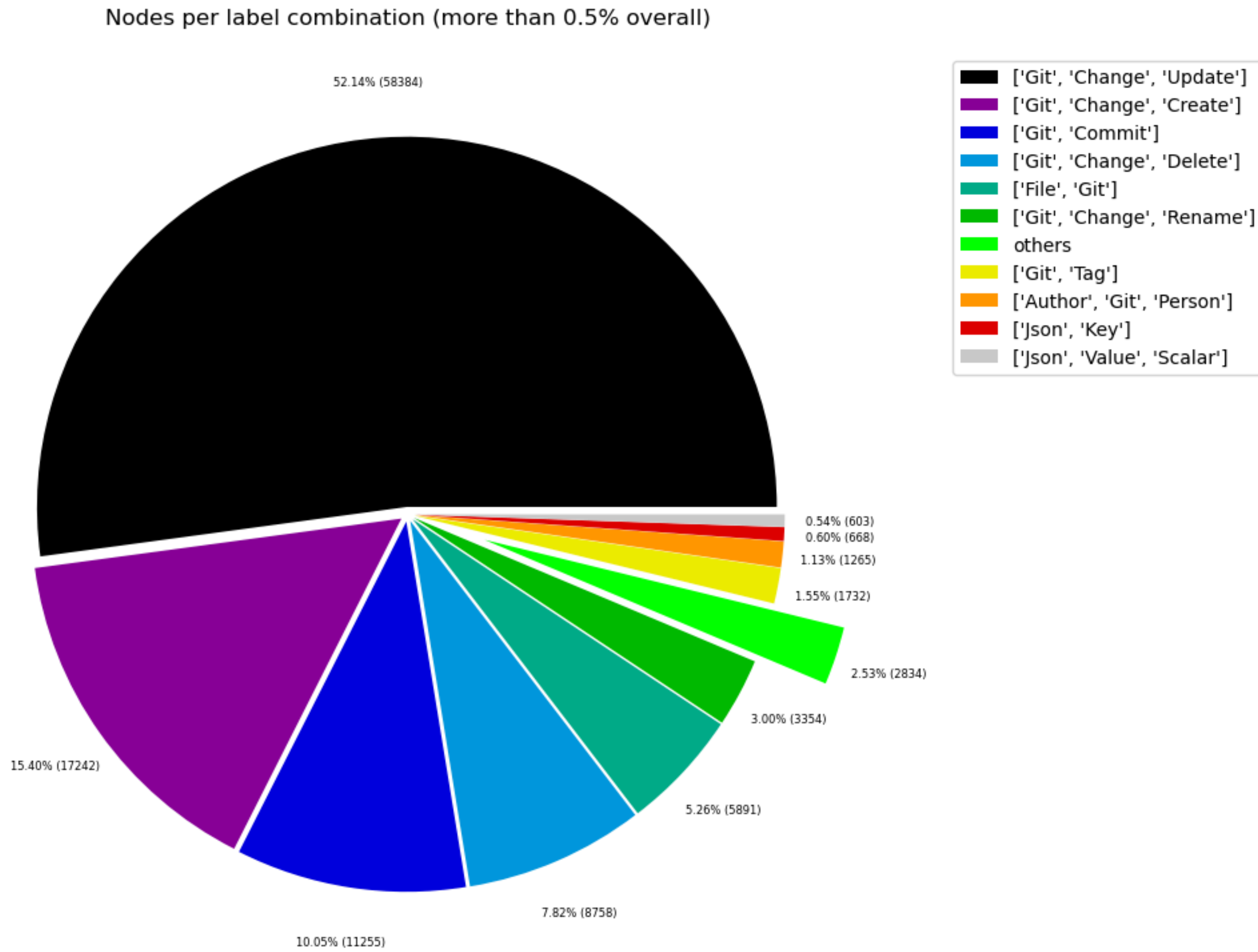


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000893
1	[NPM, PublishConfig]	1	0.000893
2	[File, TS, Scan]	1	0.000893
3	[TS, Method]	1	0.000893
4	[Repository, File, Git]	1	0.000893
5	[TS, Constructor]	1	0.000893
6	[Value, TS, ObjectMember]	1	0.000893
7	[TS, Class]	1	0.000893
8	[TS, Enum]	2	0.001786
9	[Value, Object, TS]	3	0.002679
10	[Type, TS, Tuple]	3	0.002679
11	[Value, TS, Function]	4	0.003572
12	[TS, TypeParameter]	4	0.003572
13	[Value, TS, Complex]	5	0.004465
14	[Repository, NPM]	5	0.004465
15	[NPM, Engine]	6	0.005358
16	[Project, TS]	6	0.005358
17	[File, Local]	6	0.005358
18	[Type, TS, TypeParameterReference]	6	0.005358
19	[Value, TS, Member]	6	0.005358
20	[File, TS, Local, Module]	6	0.005358
21	[Value, TS, Call]	6	0.005358
22	[TS, EnumMember]	8	0.007144
23	[Type, TS, NotIdentified]	11	0.009823
24	[TS, ExternalModule]	11	0.009823
25	[Json, Value, Array]	12	0.010716
26	[Value, TS, Declared]	13	0.011609
27	[TS, TypeAlias]	16	0.014288
28	[File, Directory, Local]	16	0.014288
29	[TS, Interface]	17	0.015180

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
```

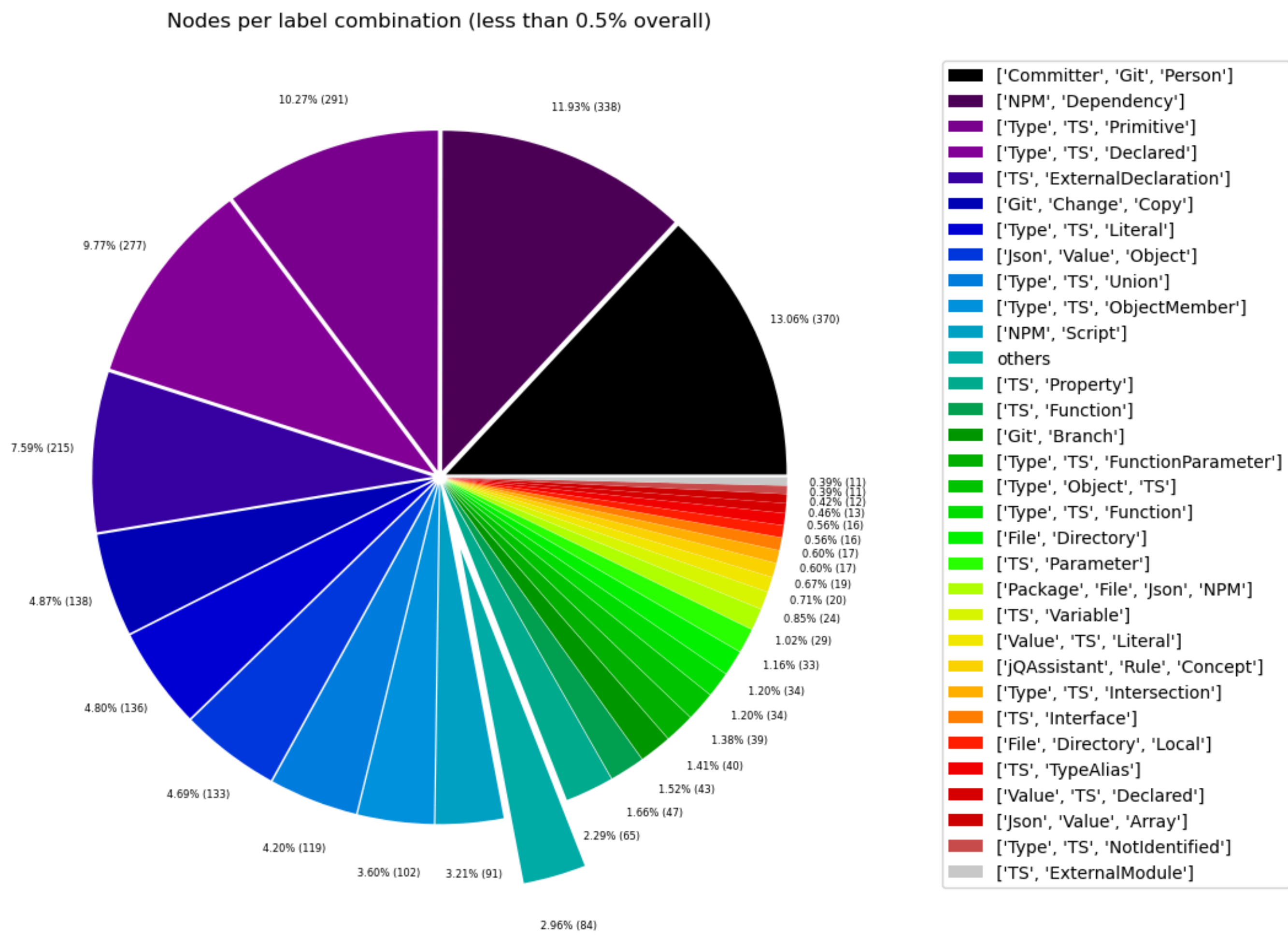


Table 1c - Highest node count by single label

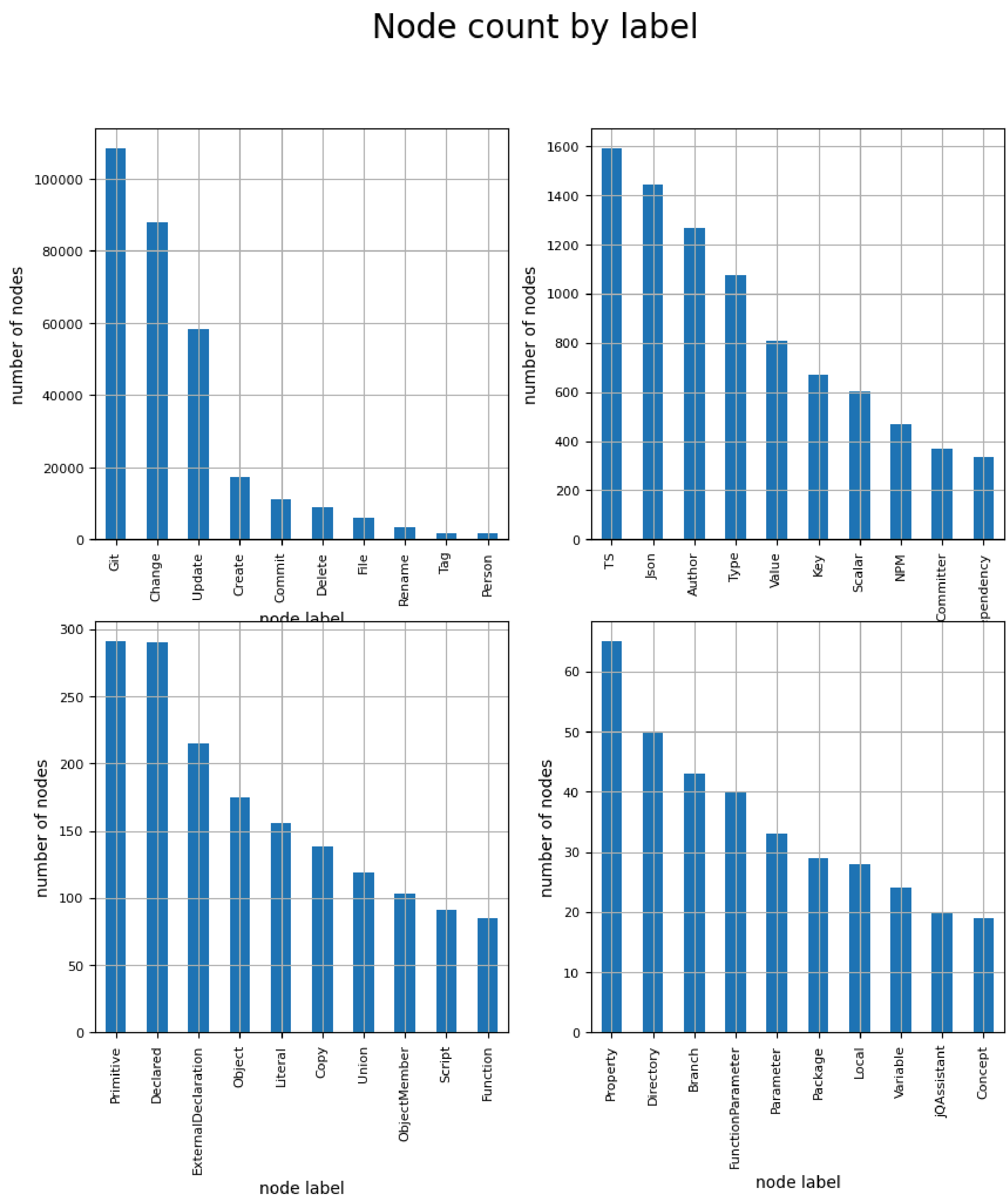
Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	108433	96.827282
1	Change	87876	78.470523
2	Update	58384	52.135088
3	Create	17242	15.396567
4	Commit	11255	10.050363
5	Delete	8758	7.820620
6	File	5984	5.343525
7	Rename	3354	2.995017
8	Tag	1732	1.546622
9	Person	1635	1.460004
10	TS	1591	1.420713
11	Json	1445	1.290340
12	Author	1265	1.129605
13	Type	1075	0.959941
14	Value	806	0.719733
15	Key	668	0.596503
16	Scalar	603	0.538460
17	NPM	470	0.419695
18	Committer	370	0.330398
19	Dependency	338	0.301823
20	Primitive	291	0.259854
21	Declared	290	0.258961
22	ExternalDeclaration	215	0.191988
23	Object	175	0.156270
24	Literal	156	0.139303
25	Copy	138	0.123230
26	Union	119	0.106263
27	ObjectMember	103	0.091976
28	Script	91	0.081260
29	Function	85	0.075902
30	Property	65	0.058043
31	Directory	50	0.044648
32	Branch	43	0.038398
33	FunctionParameter	40	0.035719
34	Parameter	33	0.029468
35	Package	29	0.025896
36	Local	28	0.025003
37	Variable	24	0.021431
38	jQAssistant	20	0.017859
39	Concept	19	0.016966

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

```
<Figure size 640x480 with 0 Axes>
```



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

```
Total number of relationships: 334408
```

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	87876	26.278079
1	MODIFIES	87876	26.278079
2	UPDATES	58384	17.458912
3	COMMITTED	22510	6.731298
4	CREATES	20734	6.200211
5	HAS_PARENT	12335	3.688608
6	DELETES	12112	3.621923
7	HAS_COMMIT	11255	3.365649
8	HAS_FILE	5891	1.761621
9	RENAMES	3354	1.002966
10	HAS_NEW_NAME	1770	0.529294
11	HAS_TAG	1732	0.517930
12	ON_COMMIT	1732	0.517930
13	HAS_AUTHOR	1265	0.378280
14	DEPENDS_ON	887	0.265245
15	HAS_KEY	668	0.199756
16	HAS_VALUE	668	0.199756
17	CONTAINS	594	0.177627
18	HAS_COMMITTER	370	0.110643
19	OF_TYPE	338	0.101074
20	EXPORTS	309	0.092402
21	REFERENCES	197	0.058910
22	DECLARES	186	0.055621
23	DECLARES_DEV_DEPENDENCY	169	0.050537
24	DECLARES_DEPENDENCY	161	0.048145
25	COPIES	138	0.041267
26	HAS_MEMBER	103	0.030801
27	HAS_TYPE_ARGUMENT	94	0.028109
28	DECLARES_SCRIPT	91	0.027212
29	RETURNS	82	0.024521

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

```
<Figure size 640x480 with 0 Axes>
```

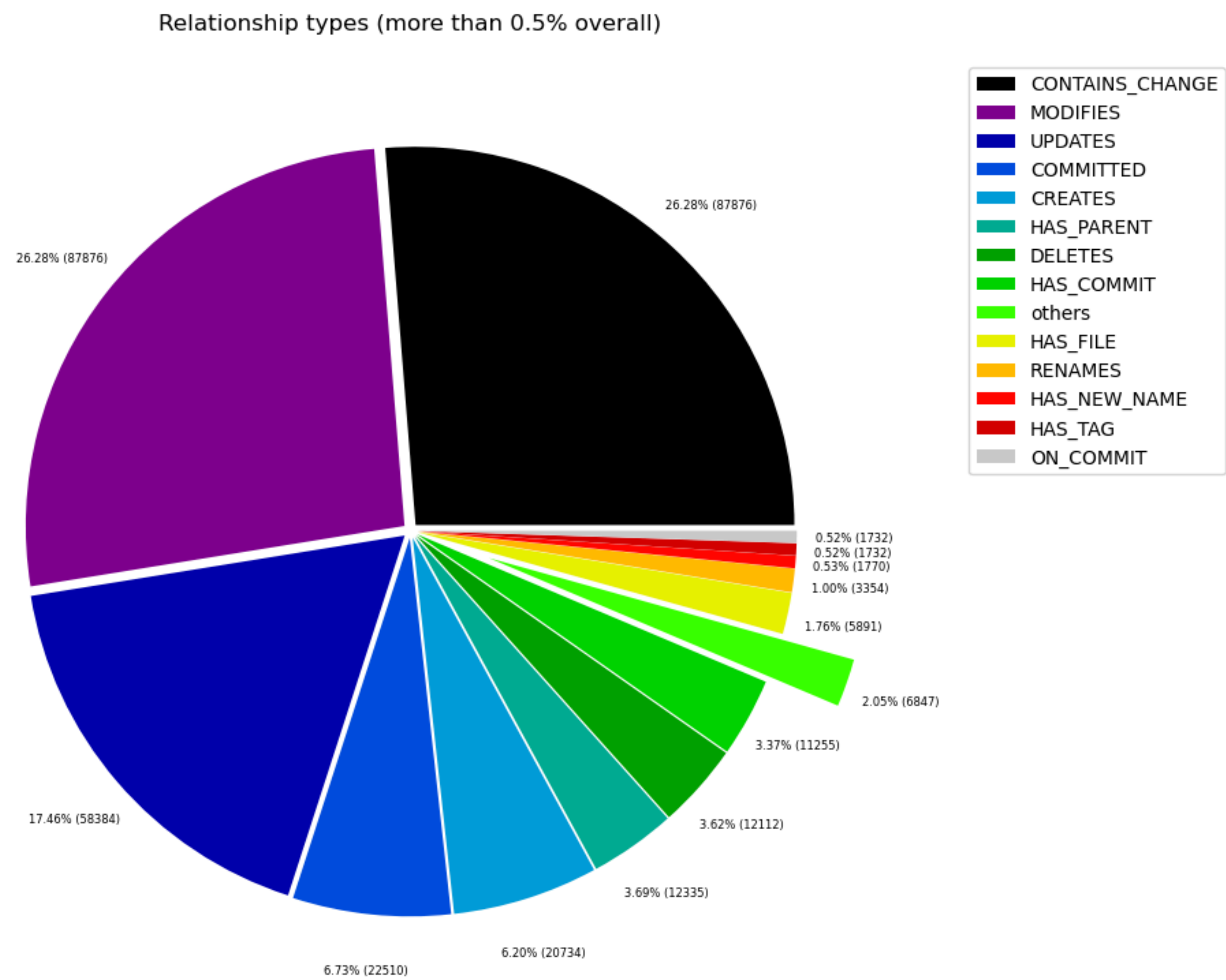


Table 2b - Lowest relationship count by type

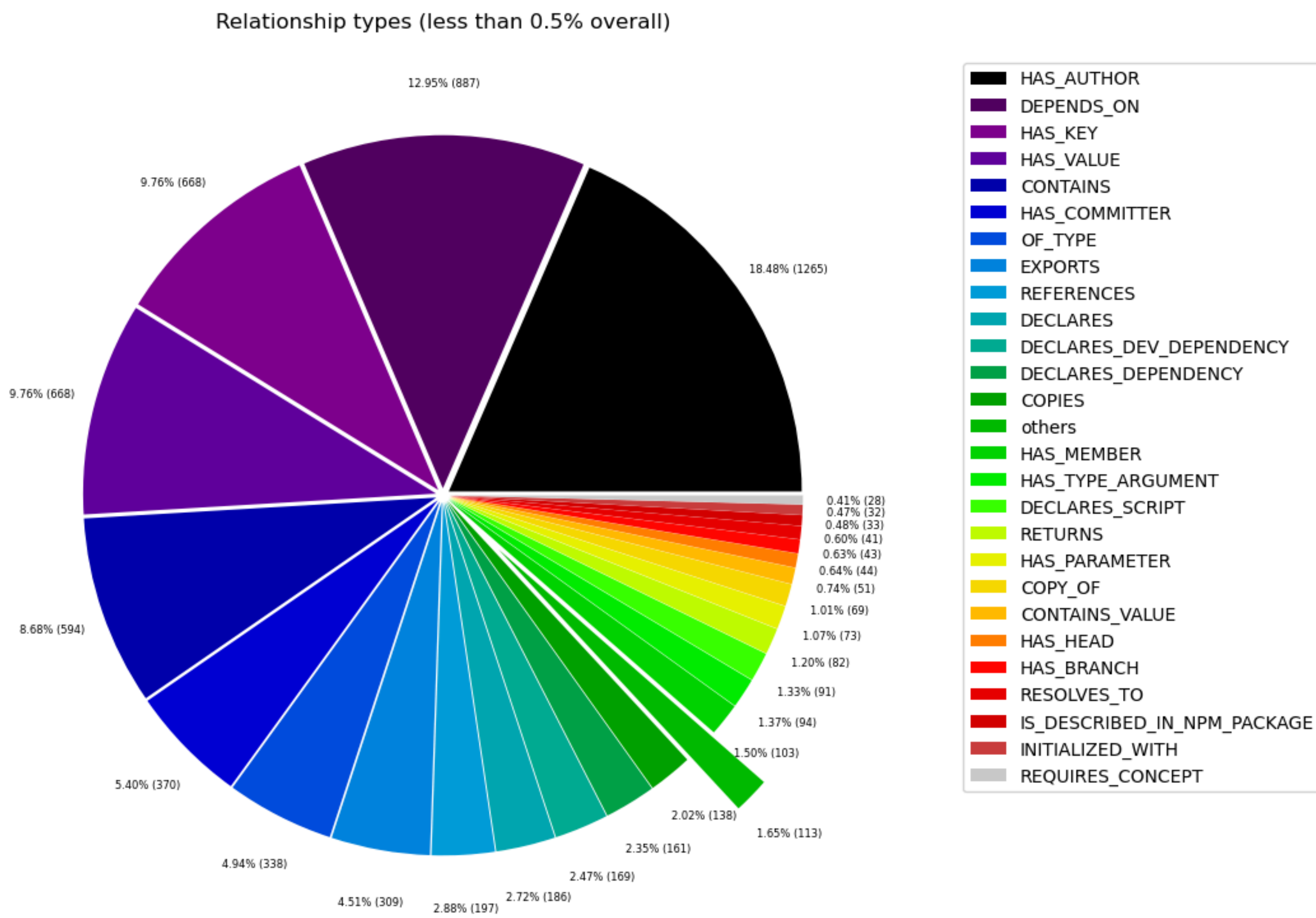
Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000299
1	CONSTRAINED_BY	4	0.001196
2	IN_REPOSITORY	5	0.001495
3	REFERENCED_PROJECTS	5	0.001495
4	PARENT	6	0.001794
5	MEMBER	6	0.001794
6	HAS_ROOT	6	0.001794
7	HAS_NPM_PACKAGE	6	0.001794
8	HAS_CONFIG	6	0.001794
9	HAS_ARGUMENT	6	0.001794
10	EXTENDS	6	0.001794
11	DECLARES_ENGINE	6	0.001794
12	CONTAINS_PROJECT	6	0.001794
13	CALLS	6	0.001794
14	DECLARES_PEER_DEPENDENCY	8	0.002392
15	USES	11	0.003289
16	INCLUDES_CONCEPT	19	0.005682
17	REQUIRES_CONCEPT	28	0.008373
18	INITIALIZED_WITH	32	0.009569
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009868
20	RESOLVES_TO	41	0.012260
21	HAS_BRANCH	43	0.012859
22	HAS_HEAD	44	0.013158
23	CONTAINS_VALUE	51	0.015251
24	COPY_OF	69	0.020633
25	HAS_PARAMETER	73	0.021830
26	RETURNS	82	0.024521
27	DECLARES_SCRIPT	91	0.027212
28	HAS_TYPE_ARGUMENT	94	0.028109
29	HAS_MEMBER	103	0.030801

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

```
<Figure size 640x480 with 0 Axes>
```



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Change, Update]	MODIFIES	[File, Git]	58384	58384	
1	[Git, Change, Update]	UPDATES	[File, Git]	58384	58384	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	58384	11255	
3	[Git, Change, Create]	MODIFIES	[File, Git]	17242	17242	
4	[Git, Change, Create]	CREATES	[File, Git]	17242	17242	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17242	11255	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12335	11255	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11255	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11255	1265	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11255	370	
10	[Git, Change, Delete]	DELETES	[File, Git]	8758	8758	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8758	8758	
12	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8758	11255	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5891	1	
14	[Git, Change, Rename]	DELETES	[File, Git]	3354	3354	
15	[Git, Change, Rename]	CREATES	[File, Git]	3354	3354	
16	[Git, Change, Rename]	RENAMES	[File, Git]	3354	3354	
17	[Git, Change, Rename]	MODIFIES	[File, Git]	3354	3354	
18	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3354	11255	
19	[File, Git]	HAS_NEW_NAME	[File, Git]	1770	5891	
20	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1732	1	
21	[Git, Tag]	ON_COMMIT	[Git, Commit]	1732	1732	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1265	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

```
total_number_of_nodes (vertices): 111986
total_number_of_relationships (edges): 334408
-> total directed graph density: 2.666570508691998e-05
-> total directed graph density in percent: 0.0026665705086919983
```